AIR CLEANER HOUSING (Continued)

### INSTALLATION

(1) Install the air cleaner housing mounting bracket.
(2) Install the air cleaner housing and mounting nut. Tighten nut to 4.5 N·m ( 40 in. lbs.).
(3) Install vent hose.
(4) Install air cleaner element into the housing.
(5) Position the housing cover into the locating tabs on the housing and install the cover.
(6) Pry up the spring clips and lock the cover in place.
(7) Install the air intake duct. (Fig. 6)
(8) Install the air outlet duct.

## CYLINDER HEAD

### DESCRIPTION

The cross flow designed, aluminum cylinder head contains dual over-head camshafts with four valves per cylinder (Fig. 7). The valves are arrange in two in-line banks. The intake valves face toward the left side of the vehicle. The exhaust valves face the right side. The cylinder head incorporates powdered metal valve guides and seats. The cylinder head is sealed to the block using a multi-layer steel head gasket and retaining bolts.

Integral oil galleries providing lubrication passages to the hydraulic lash adjusters, camshafts, and valve mechanisms.

### OPERATION

The cylinder head closes the combustion chamber, allowing the pistons to compress the fuel/air mixture for ignition. The valves are actuated by the lobe profiles on the camshaft to open and close at specified duration to either allow clean air in the combustion chamber or the exhaust gases out; depending on the stroke of the engine.

## DIAGNOSIS AND TESTING

### CYLINDER HEAD GASKET

A cylinder head gasket leak can be located between adjacent cylinders or between a cylinder and the adjacent water jacket.

Possible indications of the cylinder head gasket leaking between adjacent cylinders are:

- Loss of engine power
- Engine misfiring
- Poor fuel economy

Possible indications of the cylinder head gasket leaking between a cylinder and an adjacent water jacket are:

- Engine overheating

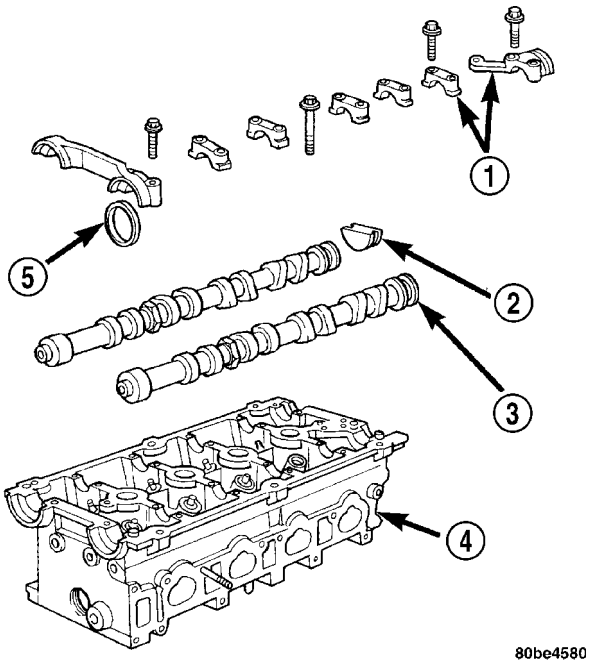


*Fig. 7 Cylinder Head and Camshafts*

1 - CAMSHAFT BEARING CAPS
2 - PLUG
3 - CAMSHAFT
4 - CYLINDER HEAD
5 - CAMSHAFT OIL SEAL

- Loss of coolant
- Excessive steam (white smoke) emitting from exhaust
- Coolant foaming

### CYLINDER-TO-CYLINDER LEAKAGE TEST

To determine if an engine cylinder head gasket is leaking between adjacent cylinders, follow the procedures in Cylinder Compression Pressure Test (Refer to 9 - ENGINE - DIAGNOSIS AND TESTING). An engine cylinder head gasket leaking between adjacent cylinders will result in approximately a 50 - 70% reduction in compression pressure.

### CYLINDER-TO-WATER JACKET LEAKAGE TEST

**WARNING: USE EXTREME CAUTION WHEN THE ENGINE IS OPERATING WITH COOLANT PRESSURE CAP REMOVED.**

#### VISUAL TEST METHOD

With the engine cool, remove the coolant pressure cap. Start the engine and allow it to warm up until thermostat opens.

If a large combustion/compression pressure leak exists, bubbles will be visible in the coolant.